Relevant passage ranking

Alpha Follows the Hops

Query Complexity and Score Fusion in Graph-Augmented Re-ranking

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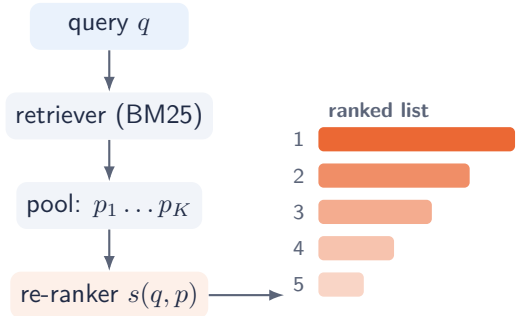


Part 1

Background

The task: rank passages for a query

- Input: a query q and a pool of candidate passages $p_1, \ldots, p_K$.
- Output: the passages ordered by relevance to q.
- A **scoring function** gives each pair (q, p_i) a score $s(q, p_i)$; the ranking sorts by it.
- Usual pipeline: a cheap retriever fills the pool (BM25, top-K), an expensive model *re-ranks* it.
- Evaluated with nDCG@10 and MRR@10.



Scoring functions: one score per pair

TF-IDF

$\sum_{t \in q} \mathrm{tf}(t, p) \cdot \mathrm{idf}(t)$

Exact term overlap, weighted by rarity.

BM25

$\sum_{t \in q} \mathrm{idf}(t) \, \dfrac{\mathrm{tf}\,(k_1 + 1)}{\mathrm{tf} + k_1 B_p}$

TF saturation and length normalisation; the standard retriever.

Bi-encoder

$\cos\big(E(q),\, E(p)\big)$

Query and passage encoded separately; fast, indexable.

Cross-encoder

$\mathrm{BERT}\big([q\,;\,p]\big)$

Joint encoding; most accurate, too slow to index.

lexical $\longrightarrow$ neural: more accurate, more expensive

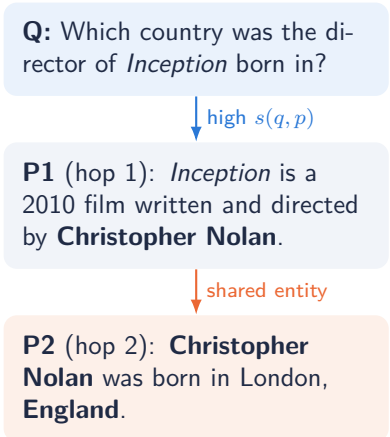
All of them score each passage **on its own**: $s(q, p_i)$ never looks at the other candidates.

Where it breaks: multi-hop questions

Single-hop: one passage answers the query.

Multi-hop: the answer needs a **chain** of passages.

- The second passage (about the director) shares almost no words with the query.
- A pointwise scorer ranks it low, although it is needed.
- Its relevance shows only *through its link* to P1.



Illustrative example.

GAR: Graph-based Adaptive Re-ranking

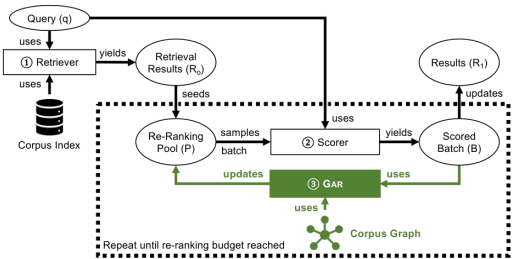


Figure 1 of MacAvaney et al. (CIKM 2022).

- **Clustering hypothesis**: passages close to a relevant passage tend to be relevant too.
- Offline: a **corpus graph**, each passage linked to its k most similar passages.
- Online: after each scored batch, the *neighbours* of the top passages join the pool.
- Fixes the recall limit of the first-stage pool; the scorer itself stays pointwise.

GAR: the re-ranking loop

Algorithm 1 Graph-based Adaptive Re-Ranking

Input: Initial ranking R_0, batch size b, budget c, corpus graph G
Output: Re-Ranked pool R_1

$R_1 \leftarrow \emptyset$ ▷ Re-Ranking results
$P \leftarrow R_0$ ▷ Re-ranking pool
$F \leftarrow \emptyset$ ▷ Graph frontier
do
  $B \leftarrow \textsc{Score}(\text{top } b \text{ from } P, \text{subject to } c)$ ▷ e.g., monoT5
  $R_1 \leftarrow R_1 \cup B$ ▷ Add batch to results
  $R_0 \leftarrow R_0 \setminus B$ ▷ Discard batch from initial ranking
  $F \leftarrow F \setminus B$ ▷ Discard batch from frontier
  $F \leftarrow F \cup (\textsc{Neighbours}(B, G) \setminus R_1)$ ▷ Update frontier
  $P \leftarrow \begin{cases} R_0 & \text{if } P = F \\ F & \text{if } P = R_0 \end{cases}$ ▷ Alternate initial ranking and frontier
while $|R_1| < c$
$R_1 \leftarrow R_1 \cup \textsc{Backfill}(R_0, R_1)$ ▷ Backfill remaining items

Algorithm 1 of MacAvaney et al. (CIKM 2022).

- Alternates between the initial ranking R_0 and the *graph frontier* F.
- Up to $+8\%$ nDCG for BM25 $\gg$ monoT5 on MS MARCO.
- The graph decides **which** passages get scored, not **how** they are scored.

Next step: let the graph change the *score* itself, so a passage is judged together with its neighbours $\Rightarrow$ ReGAT-BERT.

ReGAT-BERT: the idea

Semantic score s_{sem}

A BERT cross-encoder reads (q, p) jointly.
Strong on *direct* matches.

Contextual score s_{ctx}

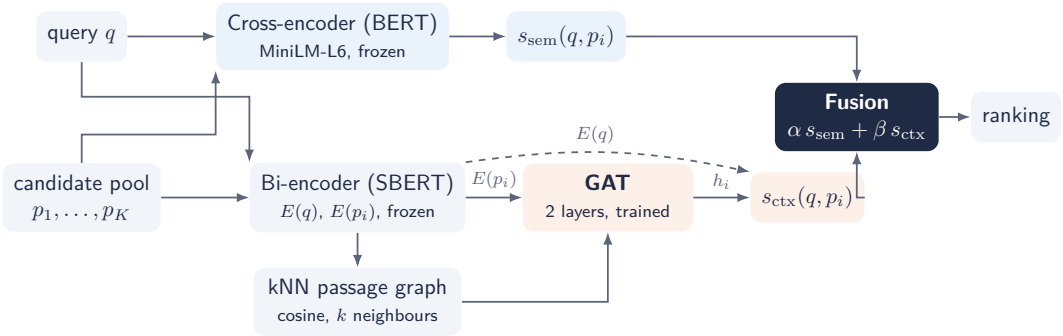
Passages form a graph; a Graph Attention Network refines each passage embedding with its neighbours.
Scores the passage *in context*.

- Two complementary views of relevance.
- Final score = a weighted sum of the two:

$$\text{score}(q, p) = \alpha\, s_{\text{sem}} + \beta\, s_{\text{ctx}}$$

- α, β are learned **once**, then the **same weights apply to every query**.

ReGAT-BERT architecture



Our diagram of the pipeline as reproduced in this project (models as used in our experiments).

Inside the GAT: attention over neighbours

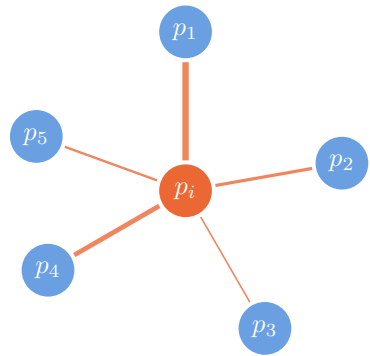
Graph: node i = passage p_i with feature $x_i = E(p_i)$; edges to its k nearest pool neighbours (cosine ≥ 0.3) and a self-loop.

Attention (GATv2, per head):

$$e_{ij} = \mathbf{a}^{\top} \,\mathrm{LeakyReLU}\big(\mathbf{W}[x_i \,\|\, x_j]\big)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \qquad x'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij}\, \mathbf{W} x_j$$

Two layers: 4 heads (concat, hidden 128) $\rightarrow$ ELU $\rightarrow$ 1 head back to dimension d.



edge width $\propto$ attention α_{ij}:
the GAT learns which neighbours matter

The contextual score: maths

1. Refine each passage with its neighbourhood (residual, so it starts at the bi-encoder embedding):

$$h_i = \frac{\mathrm{LN}\big(x_i + \mathrm{GAT}(X, A)_i\big)}{\big\|\mathrm{LN}\big(x_i + \mathrm{GAT}(X, A)_i\big)\big\|}$$

2. Score against the query vector, which lives in the same space ($h_i \in \mathbb{R}^d$, $E(q) \in \mathbb{R}^d$):

$$s_{\mathrm{ctx}}(q, p_i) = E(q)^\top h_i = \cos\big(E(q), h_i\big)$$

3. Train the GAT once (encoders frozen) with InfoNCE over the pool:

$$\mathcal{L}_{\mathrm{GAT}} = -\log \frac{\sum_{p^+} \exp(s_{\mathrm{ctx}}(q, p^+)/\tau)}{\sum_{j=1}^{K} \exp(s_{\mathrm{ctx}}(q, p_j)/\tau)}$$

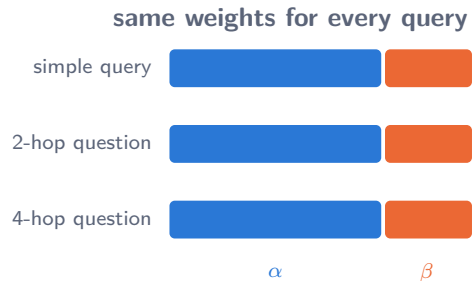
$\tau = 0.05$; positives = gold passages.

Same query vector, but a passage's vector now carries information from the passages it is linked to, e.g. the bridge entity of a 2-hop chain.

ReGAT-BERT fusion strategy

$$\text{score}(q, p) = \alpha \cdot s_{\text{sem}}(q, p) + \beta \cdot s_{\text{ctx}}(q, p)$$

- α, β: two free scalars, learned with a pointwise BCE loss on relevance labels.
- Raw scores are fused as they are: the cross-encoder logit and the cosine live on **different scales**.
- One pair (α, β) for **all queries**: an easy keyword query and a 4-hop question get the same mix.



Part 2

Our study

Problem statement

Should the weight between the semantic and the contextual score be **the same for every query**, or should it depend on the query and its passages?

Simple / single-hop queries

One passage answers it; the cross-encoder sees everything it needs.

Expect $\alpha > 0.5$ (trust s_{sem}).

Complex / multi-hop queries

Evidence is spread over linked passages; only the graph sees the links.

Expect $\alpha < 0.5$ (trust s_{ctx}).

Hypothesis: α should *fall as the number of hops rises*, so a query-dependent $\alpha(\cdot)$ should beat a single fixed α.

Our score and the normalisers

$$\mathrm{score}(q, p_i) = \alpha \cdot \tilde{s}_{\mathrm{sem}}(q, p_i) + (1 - \alpha) \cdot \tilde{s}_{\mathrm{ctx}}(q, p_i), \qquad \alpha \in (0, 1)$$

$\tilde{s}$ = score normalised *within each query's pool*, so α is a true mixing weight. $\alpha = 1$: cross-encoder only; $\alpha = 0$: GAT only.

None

$\tilde{s}_i = s_i$

Raw scores (paper setting).

Min–max

$\tilde{s}_i = \frac{s_i - \min s}{\max s - \min s}$

Maps to $[0, 1]$; our primary normaliser.

Z-score

$\tilde{s}_i = \frac{s_i - \mu_q}{\sigma_q}$

Zero mean, unit variance per query.

Rank

$\tilde{s}_i = 1 - \frac{\mathrm{rank}_i - 1}{K - 1}$

Only the order counts; scale-free.

Modelling the dependency of α

Gate: a small MLP whose **input** decides what α may depend on:

$$\alpha(x) = \epsilon + (1-2\epsilon)\,\sigma\big(\mathbf{w}_2^\top \mathrm{ReLU}(\mathbf{W}_1 x + \mathbf{b}_1) + b_2\big)$$

Gate	input x
fixed	none: one learned scalar
Q	query embedding $E(q) \in \mathbb{R}^{384}$
S	12 score statistics $f(s_{\text{sem}}, s_{\text{ctx}})$
QS	$[E(q) \,\|\, f]$
QS+C	$[E(q) \,\|\, f \,\|$ len, #entities$]$

S features (per query, from raw scores, never labels): for each of s_{sem}, s_{ctx}: mean, std, top-1, top-1−top-2 gap, softmax entropy; plus Spearman$(s_{\text{sem}}, s_{\text{ctx}})$ and top-10 overlap.

Training (CE and GAT frozen):

$$\mathcal{L} = -\log \sum_{p^+} \mathrm{softmax}_i\Big(\tfrac{\mathrm{score}_i}{T}\Big) + \lambda\,(\alpha - \alpha_0)^2$$

b_2 starts at $\mathrm{logit}(\alpha_0)$ (α_0 = best fixed α on val): every gate starts as the fixed baseline. $\lambda = 0.01$, 5 seeds.

Datasets: from single-hop to multi-hop

MS MARCO
single-hop
Real Bing queries; 1 relevant passage per query.

HotpotQA
2-hop
Bridge and comparison questions.

BEIR-HQA
2-hop, open
HotpotQA queries over 5M Wikipedia passages.

2Wiki
2–4 hops
4 types: comparison, compositional, inference, bridge-comparison.

MuSiQue
2, 3, 4 hops
Built by composing single-hop questions; hops labelled.

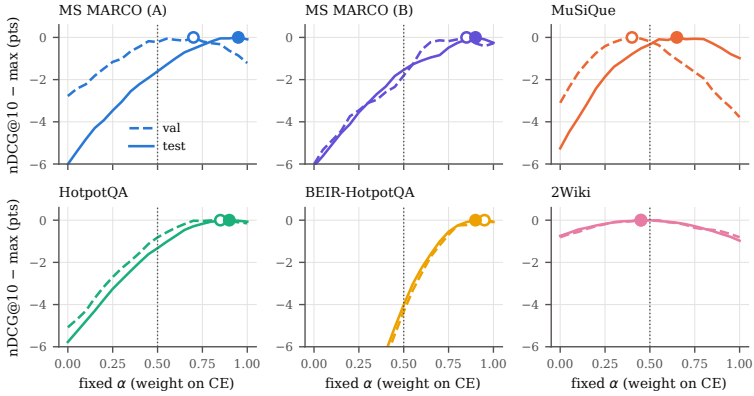
Dataset	Setting	K	GAT train	Fusion train	Val	Test	Gold/pool	Complexity labels
MS MARCO (A)	open (BM25)	100	30,000	10,000	2,000	6,980	–	proxies only
MS MARCO (B)	open (BM25)	100	1,870	1,150	698	1,745	–	proxies only
MuSiQue	closed pool	20	10,000	7,000	2,938	2,417	2.65	hops 2/3/4
HotpotQA	closed pool	10	20,000	10,000	2,000	7,405	2.00	bridge/comparison
BEIR-HotpotQA	open (BM25)	100	20,000	10,000	2,000	7,405	1.59	bridge/comparison
2Wiki	closed pool	10	20,000	10,000	2,000	12,576	2.44	4 types

Table 1 of our report. Open: BM25 pool from the full corpus. Closed: the dataset's own paragraphs.

Part 3

Results

1. The best α differs by dataset



nDCG@10 vs. fixed α. MS MARCO peaks at a high α (**0.70–0.95**); the multi-hop sets MuSiQue (val **0.40**) and 2Wiki (**0.45**) peak **below 0.5**.

Figure 1 of our report (min–max, dashed = val, solid = test).

1. More hops ⇒ lower α (MuSiQue)

		nDCG@10 of each signal alone				Best fixed α						% prefer ctx	
		val		test		val			test				
Hops	n (test)	CE	GAT	CE	GAT	mm	z	rank	mm	z	rank	val	test
2-hop	1,252	76.9	76.8	73.8	68.0	**0.40**	0.50	**0.45**	0.80	0.80	0.95	48	40
3-hop	760	69.2	71.0	71.8	66.7	**0.40**	**0.40**	**0.30**	0.55	0.65	0.85	54	39
4-hop	405	66.1	71.7	61.3	63.3	**0.25**	**0.30**	**0.00**	**0.40**	**0.40**	**0.00**	60	52

Best α falls with each hop under every normaliser. Test, min–max: **0.80** → **0.55** → **0.40**; val: 0.40 → 0.40 → 0.25.

Why: from 2 to 4 hops the cross-encoder drops 12.5 nDCG points (test), the GAT only 4.7. At 4 hops the GAT is ahead.

Table 3 of our report. mm = min–max, z = z-score.

1. The trend holds under every normaliser

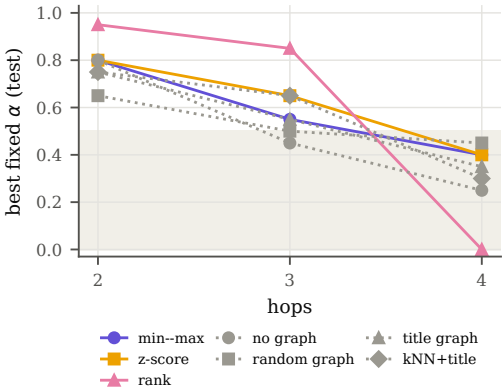


Figure 3 of our report (MuSiQue test).

- Min–max, z-score and rank all give a **monotone decline** of the best α from 2 to 4 hops.
- 4-hop questions sit **below 0.5** in all 14 normaliser / graph / split settings.
- Per question: preferred α correlates negatively with hop count (Spearman $\rho \approx -0.07$, $p \leq 0.004$ on val and test).

1. Same pattern by question type (2Wiki)

	nDCG@10 (test)		Best fixed α		% prefer	Learned Q gate: mean α and ΔnDCG@10 vs. fixed (test)					
Type	CE	GAT	val	test	ctx	min–max		z-score		rank	
comparison	97.3	95.5	0.90	0.65	30	0.72	+0.10	0.52	-0.04	0.59	-0.07
compositional	88.5	88.9	**0.45**	0.50	53	0.45	-0.01	0.34	-0.08	0.44	+0.25▲
inference	86.7	87.6	**0.25**	**0.45**	59	0.30	-0.23▽	0.22	-0.03	0.38	+0.27
bridge-comparison	90.1	92.0	**0.25**	**0.20**	67	0.35	+0.16▲	0.33	+0.22▲	0.17	+0.81▲

Comparison:
$\alpha = 0.65$–0.90,
cross-encoder wins.

Inference, bridge-comp.:
$\alpha = 0.20$–0.45,
GAT wins.

Q gate, blind to the type:
highest α on comparison,
gains on bridge-comp.

Table 4 of our report. Bold: $\alpha < 0.5$. ▲: significantly better than fixed α ($p < 0.05$).

2. Adaptive α gains on complex queries

Norm.	Gate	MSM-A	MSM-B	MuSiQue	HotpotQA	BEIR-HQA	2Wiki
none	Q	-0.48$^{\nabla}$	+0.01	+0.00	-0.00	-0.00	+0.00
	QS	-0.48$^{\nabla}$	+0.00	+0.00	+0.01	-0.00$^{\nabla}$	+0.00
	S	-0.49$^{\nabla}$	+0.00	+0.02	+0.01	-0.00	-0.00
minmax	Q	**+0.00**$^{\blacktriangle}$	+0.00	+0.00	-0.02	-0.00	+0.03
	QS	+0.00	+0.01	-0.01	+0.02	+0.00	-0.12$^{\nabla}$
	S	+0.00	**+0.02**$^{\blacktriangle}$	-0.00	+0.04	+0.01	+0.00
	QS+C	–	–	+0.00	+0.03	-0.00	-0.11$^{\nabla}$
zscore	Q	+0.01	**+0.05**$^{\blacktriangle}$	-0.15$^{\nabla}$	+0.06	**+0.18**$^{\blacktriangle}$	+0.00
	QS	**+0.02**$^{\blacktriangle}$	-0.01	-0.14$^{\nabla}$	+0.00	-0.01$^{\nabla}$	-0.00
	S	+0.01	+0.01	-0.12$^{\nabla}$	+0.01	-0.01	-0.01$^{\nabla}$
rank	Q	+0.01	+0.02	**+0.43**$^{\blacktriangle}$	-0.01	-0.01	**+0.30**$^{\blacktriangle}$
	QS	-0.12	-0.04	**+1.22**$^{\blacktriangle}$	-0.04	-0.04$^{\nabla}$	**+0.41**$^{\blacktriangle}$
	S	-0.10	-0.02	**+1.37**$^{\blacktriangle}$	-0.00	-0.02$^{\nabla}$	**+0.19**$^{\blacktriangle}$

Table 6 of our report: ΔnDCG@10 vs. best fixed α (test).

- **7 of the 11** significant gains are on the **multi-hop** sets.
- Largest: MuSiQue, rank norm., **+0.43 / +1.22 / +1.37** (Q / QS / S).
- MS MARCO gains ≤ 0.05: one fixed α serves simple queries.
- Oracle per-query α headroom: 7.4 (MuSiQue) vs. 5.0 (MS MARCO).

2. Transfer: α follows the target data

			Transferred gate, test nDCG@10				In-domain		Q gate
Setting	Train	Test	fixed	Q	QS	S	fixed	α_0	$\bar{\alpha}$ (sd)
leave-one-out	MSM+HQA	MuSiQue	71.99	71.93	71.98	72.00	71.28	0.70	0.73 (0.02)
mixed	MSM+MuSiQue+HQA	MS MARCO (A)	40.13	40.29	40.18	40.22	40.89	0.55	0.59 (0.12)
mixed	MSM+MuSiQue+HQA	HotpotQA	88.88	89.44	89.32	89.34	89.89	0.55	0.68 (0.15)
mixed	MSM+MuSiQue+HQA	MuSiQue	71.98	71.77	71.81	71.79	71.28	0.55	0.54 (0.16)
zero-shot	MSM	BEIR-HotpotQA	71.45	71.45	71.45	71.46	72.63	0.70	0.70 (0.00)
zero-shot	MSM	HotpotQA	89.24	89.24	89.25	89.20	89.89	0.70	0.70 (0.00)
zero-shot	MSM	2Wiki	90.99	90.99	90.99	90.99	91.75	0.70	0.70 (0.00)

One gate on mixed data (MS MARCO + MuSiQue + HotpotQA) gives MuSiQue the **lowest** mean α (0.54) vs. MS MARCO 0.59 and HotpotQA 0.68.

Zero-shot MS MARCO gates keep $\alpha = 0.70$ and lose to the target's own setting, e.g. 2Wiki 90.99 vs. 91.75 (in-domain $\alpha_0 = 0.45$).

Table 9 of our report (min–max, test nDCG@10 $\times 100$, 5 seeds).

Conclusion

- **Alpha follows the hops**: the more reasoning steps a query needs, the more weight belongs on the graph-contextual score.
- MuSiQue: best α 0.80 $\rightarrow$ 0.55 $\rightarrow$ 0.40 for 2 $\rightarrow$ 3 $\rightarrow$ 4 hops; 4-hop below 0.5 in every setting.
- Mechanism: the cross-encoder degrades with hops much faster than the GAT.
- One fixed weight is a compromise; query-dependent gates gain mostly on multi-hop data.

Limits

HotpotQA keeps a high α (single-hop shortcuts). Min–max gates mostly stay at their start value.

Next

Gates with explicit complexity signals (predicted hop count), and fusing at the passage-pair level.

Thank you

Questions?

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Brody, Alon, Yahav. How Attentive are Graph Attention Networks? ICLR 2022. Trivedi et al. MuSiQue. TACL 2022.

Yang et al. HotpotQA. EMNLP 2018. Ho et al. 2WikiMultiHopQA. COLING 2020. Nguyen et al. MS MARCO. 2016. Thakur et al. BEIR. NeurIPS 2021.